6. Which of the following is not a stable sorting algorithm?

(A) Insertion sort (B) Selection sort (C) Bubble sort (D) Merge sort

7. Time complexity of bubble sort in best case is

(A) θ (n) (B) θ (nlogn) (C) θ (n2) (D) θ (n(logn) 2)

8. The Θ notation in asymptotic evaluation represents

(A) Base case (B) Average case (C) Worst case (D) NULL case

9. If a problem can be broken into subproblems which are reused several times, the problem possesses which property?

(A) Overlapping subproblems (B) Optimal substructure

(C) Memoization (D) Greedy

10. In dynamic programming, the technique of storing the previously calculated values is called ?

(A) Saving value property (B) Storing value property

(C) Memoization (D) Mapping

Part - II

Answer the following questions **(2 Marks)**

1. What is an Algorithm?
2. Define Pseudo code.
3. Who is an Algorist?
4. What is Sorting?
5. What is searching? Write its types.

Part - III

Answer the following questions **(3 Marks)**

1. List the characteristics of an algorithm.
2. Discuss about Algorithmic complexity and its types.
3. What are the factors that influence time and space complexity.
4. Write a note on Asymptotic notation.
5. What do you understand by Dynamic programming?